MID TERM EXAMINATION

Course: Data Structures (CS101)

Set: A | Time: 3 Hours | Max Marks: 40 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQs

1. Which of the following is NOT typically considered a primary characteristic or benefit of a well-designed Database Management System (DBMS)? **[2 Marks]**

a) Reduced data redundancy
b) Enhanced data security
c) Increased data inconsistency
d) Improved data sharing

2. In a relational database, a column or a set of columns that uniquely identifies each row in a table is known as a: **[2 Marks]**

a) Foreign key
b) Candidate key
c) Primary key
d) Composite key

3. Which data model represents data in a tree-like structure, where data is organized into parent-child relationships, and a child can only have one parent? **[2 Marks]**

a) Relational model
b) Network model
c) Hierarchical model
d) Object-oriented model

4. The process of organizing the columns and tables of a relational database to minimize data redundancy and improve data integrity is called: **[2 Marks]**

a) Indexing
b) Normalization
c) Query optimization
d) Concurrency control

5. Which component of a DBMS is responsible for receiving user requests (e.g., SQL queries) and converting them into an efficient execution plan for the database engine? **[2 Marks]**

a) Storage Manager
b) Transaction Manager
c) Query Processor
d) Buffer Manager

6. When multiple transactions are executing concurrently, which ACID property ensures that the outcome is equivalent to some serial execution of those transactions, preventing interferences? **[2 Marks]**

a) Atomicity
b) Consistency
c) Isolation
d) Durability

SECTION A: SHORT ANSWERS

7. Briefly explain the primary purpose of a Database Management System (DBMS). **[4 Marks]**

8. Differentiate between data redundancy and data inconsistency in a database system. **[5 Marks]**

9. List and briefly describe two significant advantages of using a DBMS over a traditional file processing system. **[6 Marks]**

10. Explain the concept of data independence in a DBMS. Differentiate between logical and physical data independence. **[7 Marks]**

11. What are the ACID properties in the context of database transactions? Briefly describe two of these properties. **[6 Marks]**

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